



Queens of the Castle

How many queens can you fit **happily** on a chessboard?



Things to include

~~Using the counters (indicate slides to be removed if no counters used) then actually marking the location of the queens~~

~~How to write grid coordinates~~

~~How the queens move~~

What the aim actually is

~~*The fact there are many variations to solutions*~~

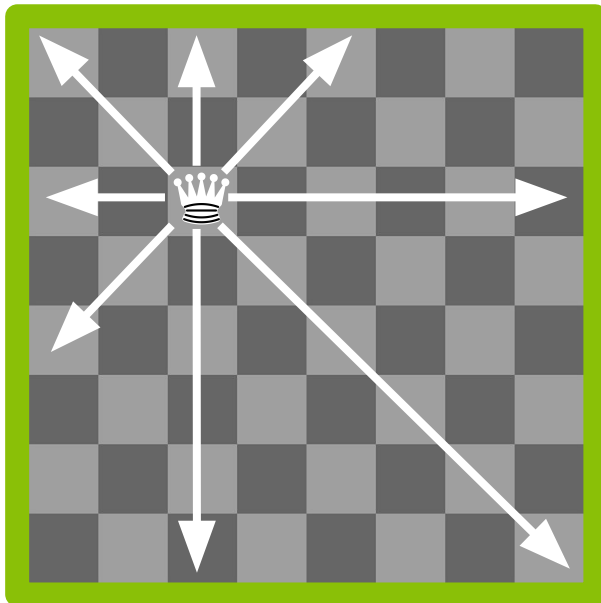
Flow of the game

humor

wrap -up;

How many unique solutions did we find? What was the most common?

Who is the Queen?



The queen is a very powerful chess piece!

She can attack in 8 directions!

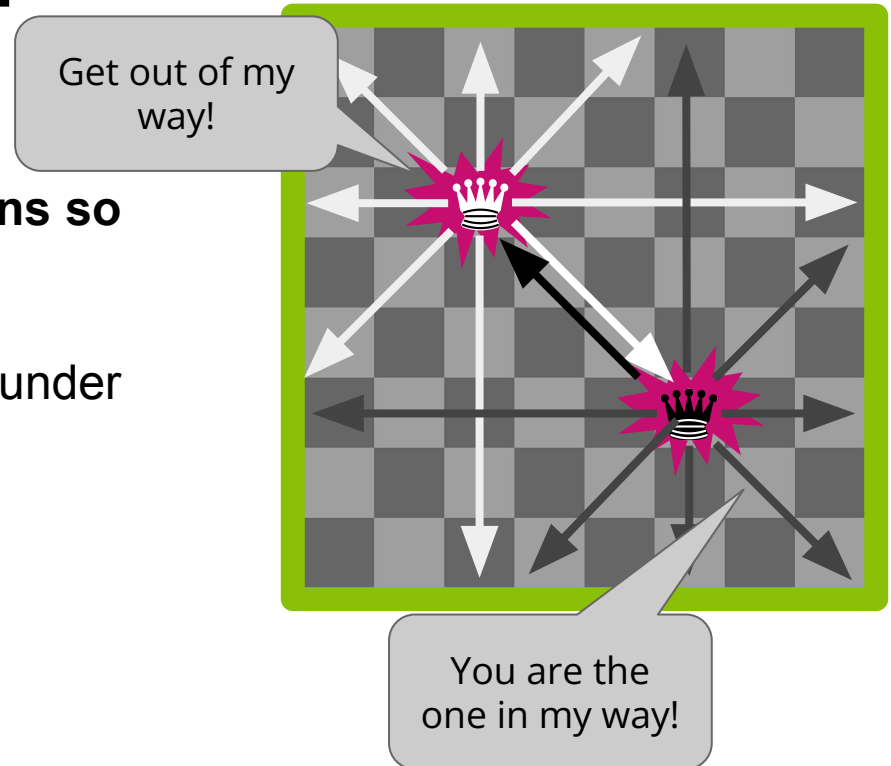
She can take on any piece in the same row or column as her. Or anything in a diagonal line to her.

More than one Queen?

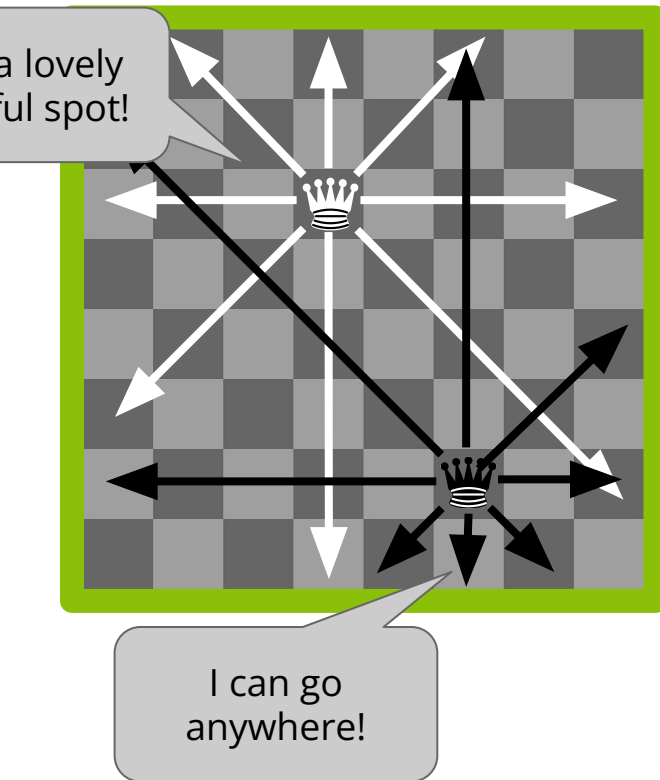
But more than one queen starts to get messy!

It's easy to accidentally place the queens so they are attacking each other.

Any piece that gets hit by an arrow is under attack!

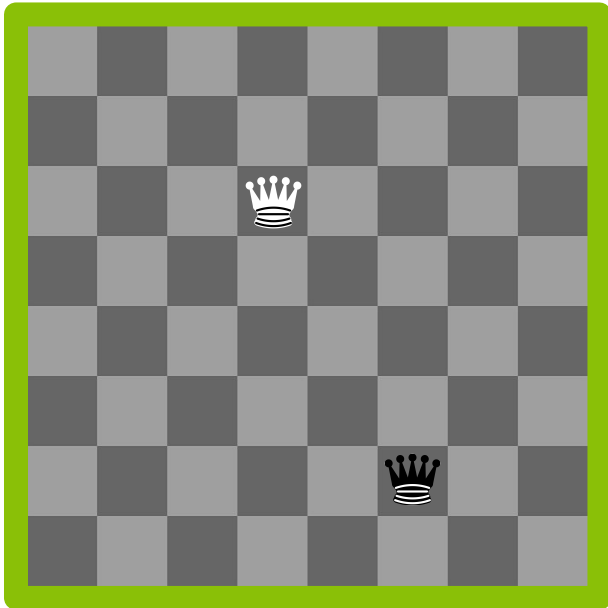


More than one Queen!



But with a little bit of effort you can make both queens happy!

Keeping Track of the Queens



I know my queens are happy now! But how do I record this solution?

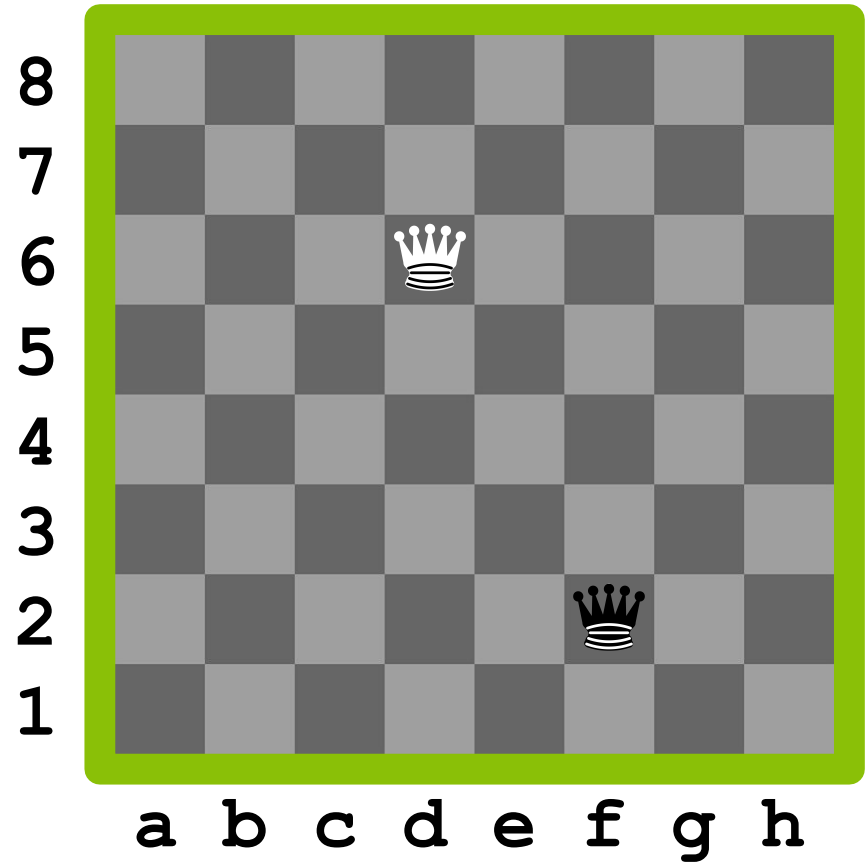
Coordinates!

Keeping Track of the Queens

Lets Label our Axis!

In chess we use numbers *and* letters!

Our x axis uses A-H,
and our y axis uses 1-8



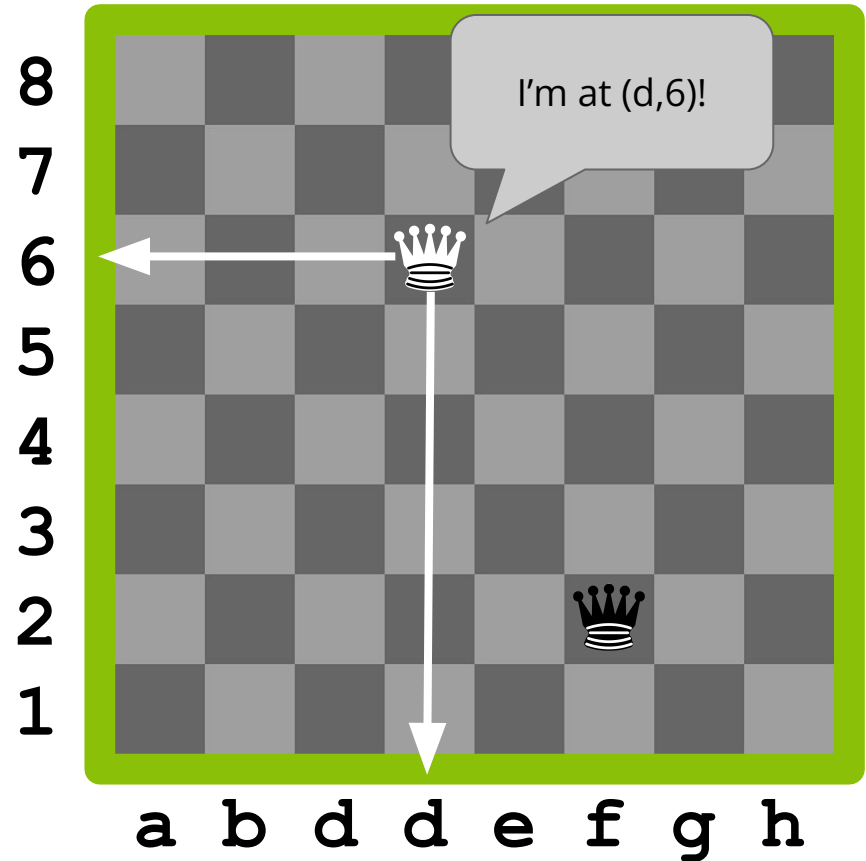
Keeping Track of the Queens

We write coordinates (x,y)

So to find the location of this queen:

1) find where it lines up on the bottom (x) axis
(d,y)

2) find where it lines up on the side (y) axis
(d,6)



Keeping Track of the Queens

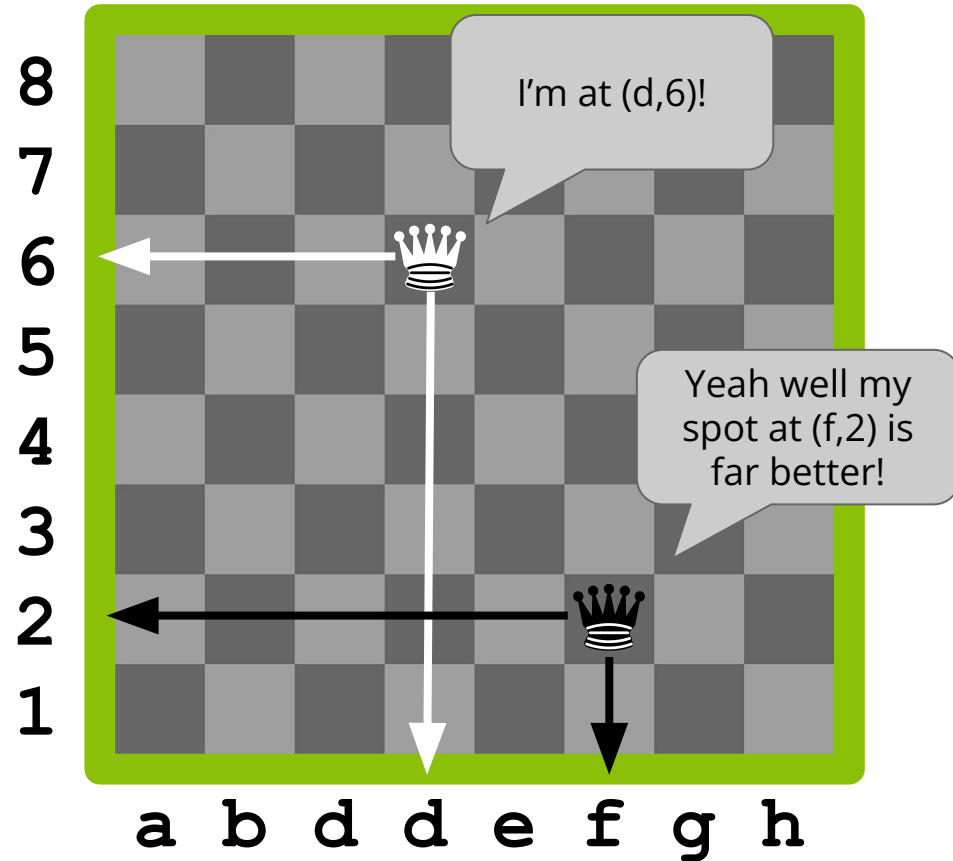
We can do this for all pieces on the board!

Then we can write out our solution for this board as:

Qd6, Qf2

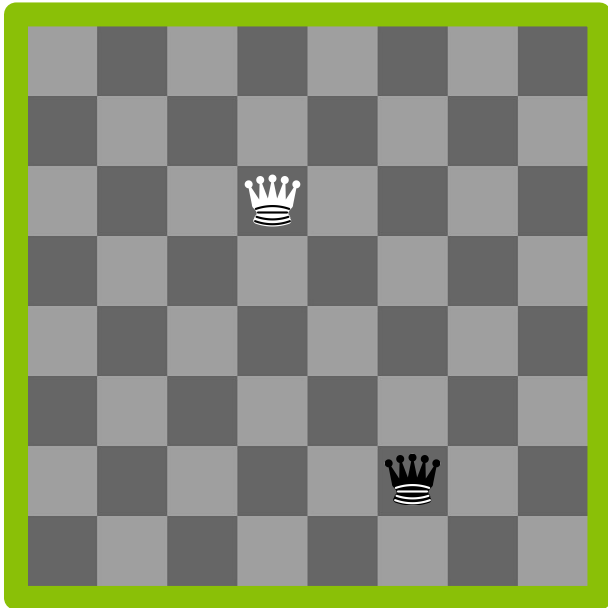
Q tells us we are using a queen piece!

This line is optional and is probably only relevant if we run a use a different piece extension activity





Queens of the Castle



Now that we know how to arrange the board to make 2 queens happy, can we add another? How many queens can we get happily sharing?

**How many queens can you get to share the board?
Can you get to 8 queens? It's very tricky!**

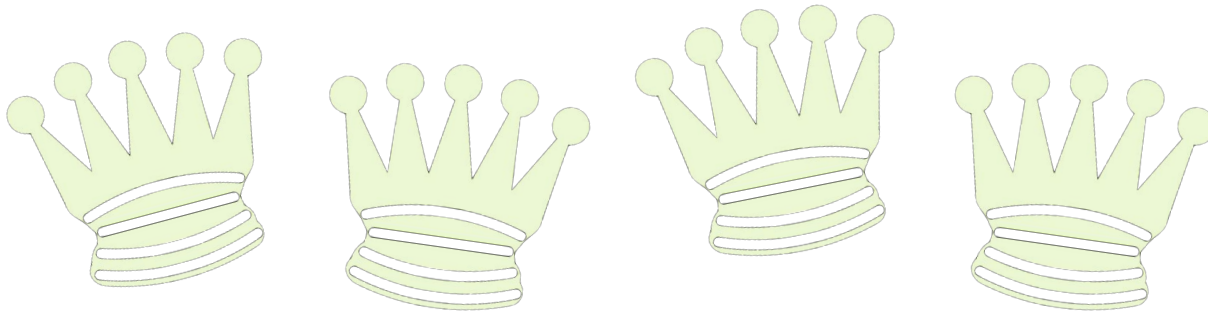




Queens of the Castle



Alright! We aren't Queen chess pieces so we should all be more than happy to group up!



Try to get into groups of 4-5 and then send 1 member up to the front to collect your materials!



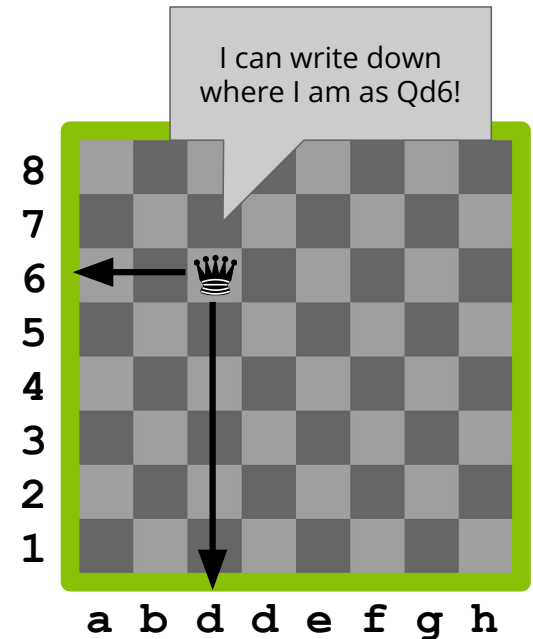
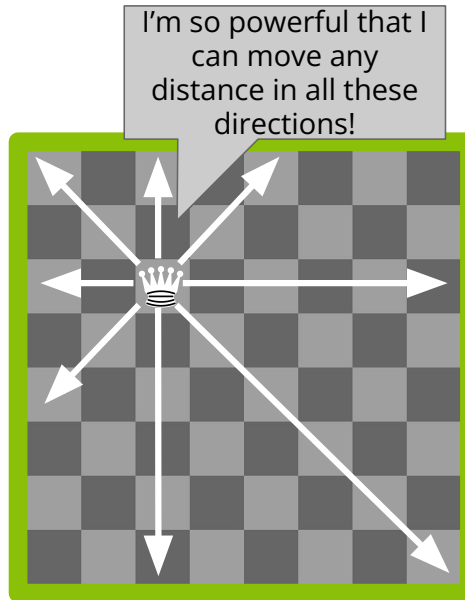


Queens of the Castle



Tip:
Use the counters to
map out your possible
solutions before
writing them down!

The tutors have extra
hints if you get stuck!



Reminder: Once you have finished the work sheet
go outside to show off your answers to the tutors!



Hint Slide

Slide tba around the idea that for some variations adding a queen can be as simple as reusing past solutions + a queen in the corner



Computerless Debrief Section

To wrap it all up together so the students understand what the point was

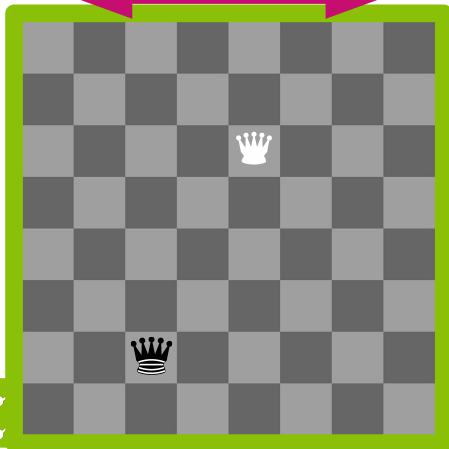
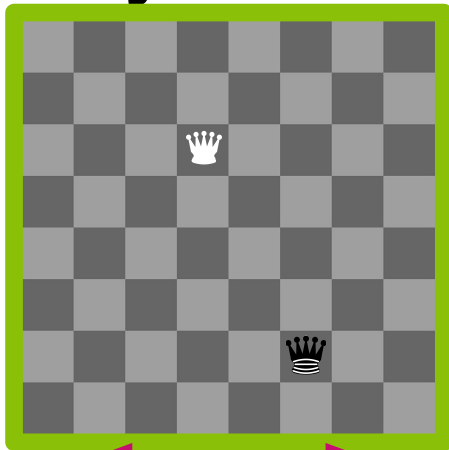




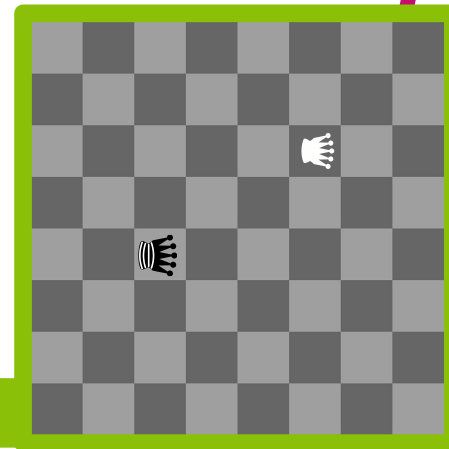
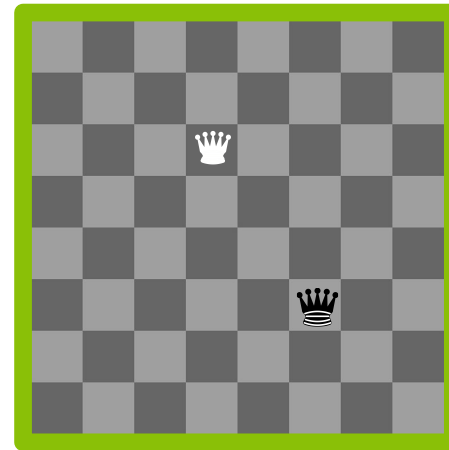
Queens of the Castle



You may have noticed there are a few different ways to solve this!



Woah! Wait a second! These solutions are the same just flipped!



And these just been

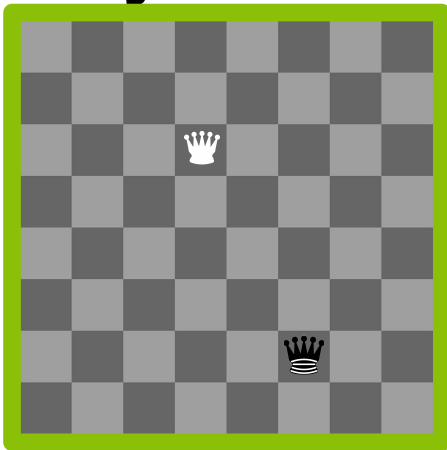




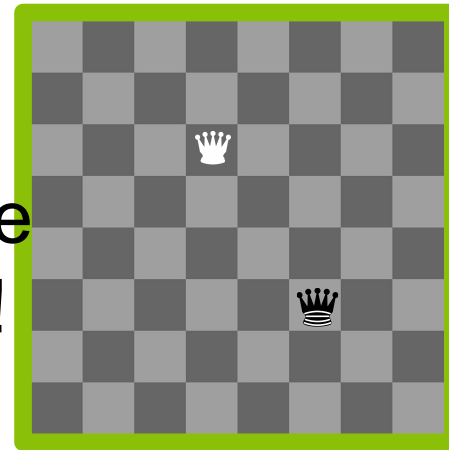
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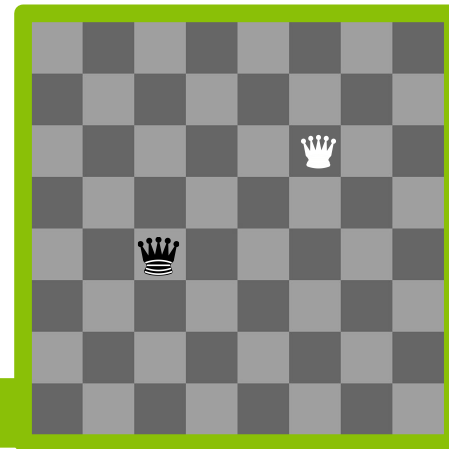
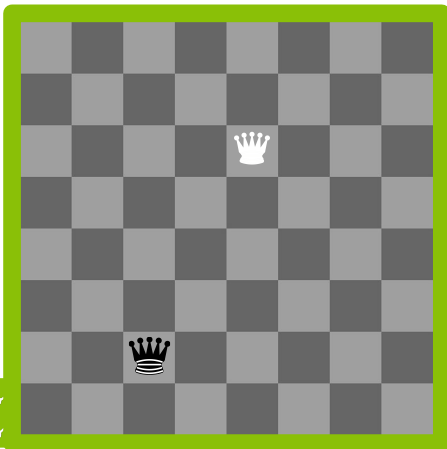
You may have noticed there are a few different ways to solve this!



This means that only these two are **unique** solutions!



this means they are non-isomorphic to each other! (but you don't need to worry about that!)



A lot of groups were able to find more than 1 unique solution, lets see how many we got overall



Let's do it!

- Get into groups of 3 - 5
- Each group gets:
 - <List of materials to pick up>
 - <List of instruction to remember>

Update timer to however long the activity should take

20:00

**Ask a tutor if
you get stuck!**



Takeaways

<Here we call out what we wanted the students to learn/take away from the activity>

- Reinforce the link to the content (if there is one)
- Summarise the Computer Science concept
- Link to real-world examples/jobs where this concept comes up

You should edit the last slide to be a bit more specific to this game, and then use this slide to put the “answers” to the questions (it’s very open ended, and there are many correct answers)



Takeaways

- What did you learn from this activity?
- How do you think what you learned could be used in a real world way?

